Pandas

October 27, 2026

1 Pandas

[1]:
```
import pandas as pd
```

[2]:
```
df1 = pd.DataFrame(
    {"a" : [4, 5, 6],
     "b" : [7, 8, 9],
     "c" : [10, 11, 12]},
     index = [1, 2, 3])
```

[3]:
```
df1
```

[3]:

	a	b	c
1	4	7	10
2	5	8	11
3	6	9	12

[4]:
```
df2 = pd.DataFrame(
    [[4, 7, 10],
     [5, 8, 11],
     [6, 9, 12]],
     index=[1, 2, 3],
     columns=['a', 'b', 'c'])
```

[5]:
```
df2
```

[5]:

	a	b	c
1	4	7	10
2	5	8	11
3	6	9	12

[6]:
```
df1["a"]
```

[6]:
```
1    4
2    5
3    6
Name: a, dtype: int64
```

[7]:
```
data = pd.Series([0.25, 0.5, 0.75, 1.0])
```

[8]:
```
data[1:2]
```

[8]:
```
1    0.5
dtype: float64
```

[9]:
```
data[0:3]
```

[9]:
```
0    0.25
1    0.50
2    0.75
dtype: float64
```

[10]:
```
data[1:]
```

[10]:
```
1    0.50
2    0.75
3    1.00
dtype: float64
```

[11]:
```
data[:2]
```

[11]:
```
0    0.25
1    0.50
dtype: float64
```

2 Using Dataset 'president_heights1.csv'

[12]:
```
df = pd.read_csv('president_heights1.csv')
```

[13]:
```
df
```

[13]:

	order	name	height(cm)
0	16	Abraham Lincoln	193.0
1	7	Andrew Jackson	185.0
2	17	Andrew Johnson	178.0
3	44	Barack Obama	NaN
4	23	Benjamin Harrison	168.0
5	42	Bill Clinton	188.0
6	30	Calvin Coolidge	178.0
7	21	Chester A. Arthur	183.0
8	34	Dwight D. Eisenhower	179.0
9	32	Franklin D. Roosevelt	188.0
10	14	Franklin Pierce	NaN
11	41	George H. W. Bush	188.0
12	43	George W. Bush	182.0

```
13      1       George Washington       189.0
14     38             Gerald Ford         NaN
15     33          Harry S. Truman       175.0
16     31           Herbert Hoover       182.0
17     20       James A. Garfield         NaN
18     15          James Buchanan       183.0
19     11           James K. Polk       173.0
20      4           James Madison       163.0
21      5            James Monroe       183.0
22     39            Jimmy Carter       177.0
23      2              John Adams       170.0
24     35          John F. Kennedy         NaN
25      6       John Quincy Adams       171.0
26     10             John Tyler       183.0
27     36       Lyndon B. Johnson       193.0
28      8        Martin Van Buren       168.0
29     13        Millard Fillmore       175.0
30     37           Richard Nixon       182.0
31     40           Ronald Reagan       185.0
32     19     Rutherford B. Hayes       174.0
33     26      Theodore Roosevelt       178.0
34      3        Thomas Jefferson       189.0
35     18        Ulysses S. Grant       173.0
36     29       Warren G. Harding       183.0
37      9  William Henry Harrison       173.0
38     27     William Howard Taft       182.0
39     25        William McKinley       170.0
40     28          Woodrow Wilson       180.0
41     12          Zachary Taylor       173.0
```

[14]:
```
df.head(5)
```

[14]:
```
   order              name  height(cm)
0     16    Abraham Lincoln       193.0
1      7     Andrew Jackson       185.0
2     17     Andrew Johnson       178.0
3     44       Barack Obama         NaN
4     23  Benjamin Harrison       168.0
```

[15]:
```
df.tail(5)
```

[15]:
```
    order                    name  height(cm)
37      9  William Henry Harrison       173.0
38     27     William Howard Taft       182.0
39     25        William McKinley       170.0
40     28          Woodrow Wilson       180.0
41     12          Zachary Taylor       173.0
```

[16]:
```
df['name'].value_counts()
```

[16]:
```
name
Abraham Lincoln           1
Andrew Jackson            1
Andrew Johnson            1
Barack Obama              1
Benjamin Harrison         1
Bill Clinton              1
Calvin Coolidge           1
Chester A. Arthur         1
Dwight D. Eisenhower      1
Franklin D. Roosevelt     1
Franklin Pierce           1
George H. W. Bush         1
George W. Bush            1
George Washington         1
Gerald Ford               1
Harry S. Truman           1
Herbert Hoover            1
James A. Garfield         1
James Buchanan            1
James K. Polk             1
James Madison             1
James Monroe              1
Jimmy Carter              1
John Adams                1
John F. Kennedy           1
John Quincy Adams         1
John Tyler                1
Lyndon B. Johnson         1
Martin Van Buren          1
Millard Fillmore          1
Richard Nixon             1
Ronald Reagan             1
Rutherford B. Hayes       1
Theodore Roosevelt        1
Thomas Jefferson          1
Ulysses S. Grant          1
Warren G. Harding         1
William Henry Harrison    1
William Howard Taft       1
William McKinley          1
Woodrow Wilson            1
Zachary Taylor            1
Name: count, dtype: int64
```

[17]:
```
len(df)
```

[17]: 42

[18]:
```
df.shape
```

[18]: (42, 3)

[19]:
```
df['name'].nunique()
```

[19]: 42

[20]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 42 entries, 0 to 41
Data columns (total 3 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   order       42 non-null     int64
 1   name        42 non-null     str
 2   height(cm)  37 non-null     float64
dtypes: float64(1), int64(1), str(1)
memory usage: 1.7 KB
```

[21]:
```
df.describe()
```

[21]:

	order	height(cm)
count	42.000000	37.000000
mean	22.476190	179.378378
std	13.152461	7.361127
min	1.000000	163.000000
25%	11.250000	173.000000
50%	22.000000	180.000000
75%	33.750000	183.000000
max	44.000000	193.000000

[22]:
```
df['height(cm)'].median()
```

[22]: np.float64(180.0)

[23]:
```
df['height(cm)'].std()
```

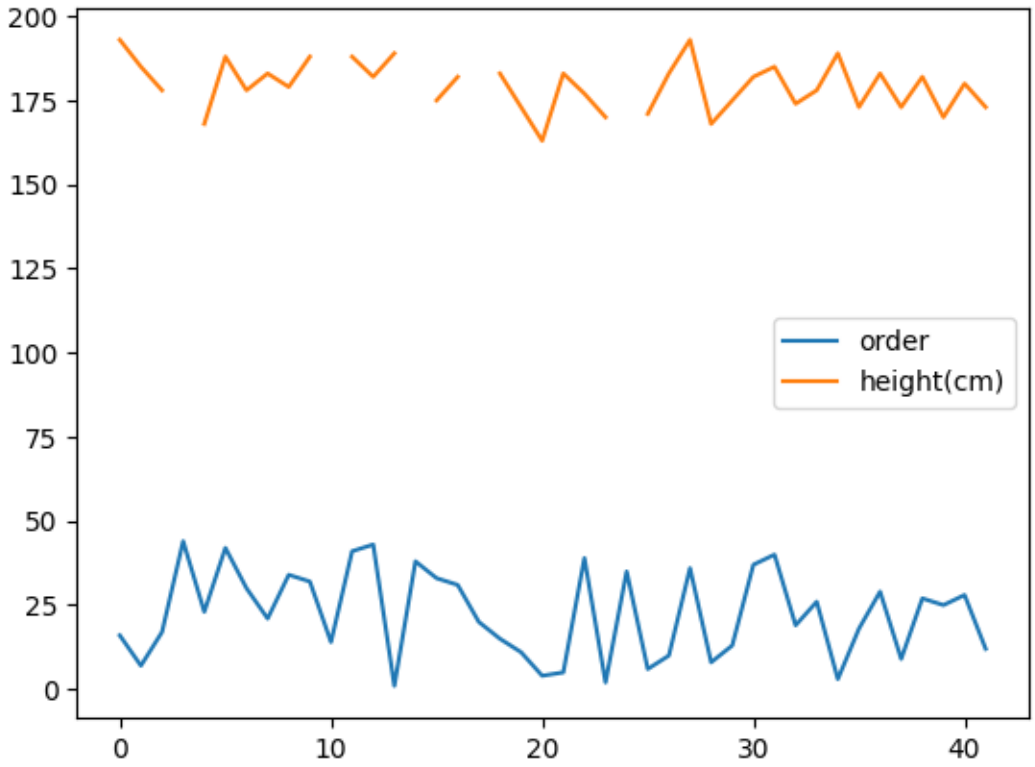
[23]: np.float64(7.361126692713975)

[24]:
```
df['height(cm)'].quantile([0.25,0.75])
```

[24]:
```
0.25    173.0
0.75    183.0
Name: height(cm), dtype: float64
```

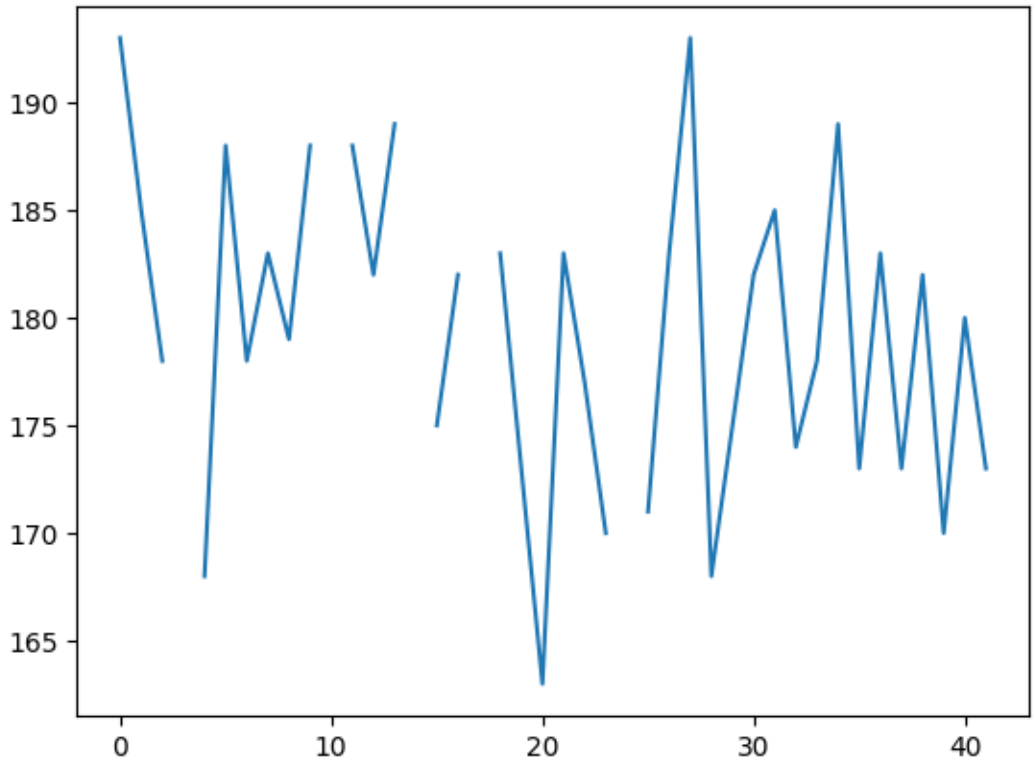
[25]:
```
df.plot()
```

[25]:
```
<Axes: >
```



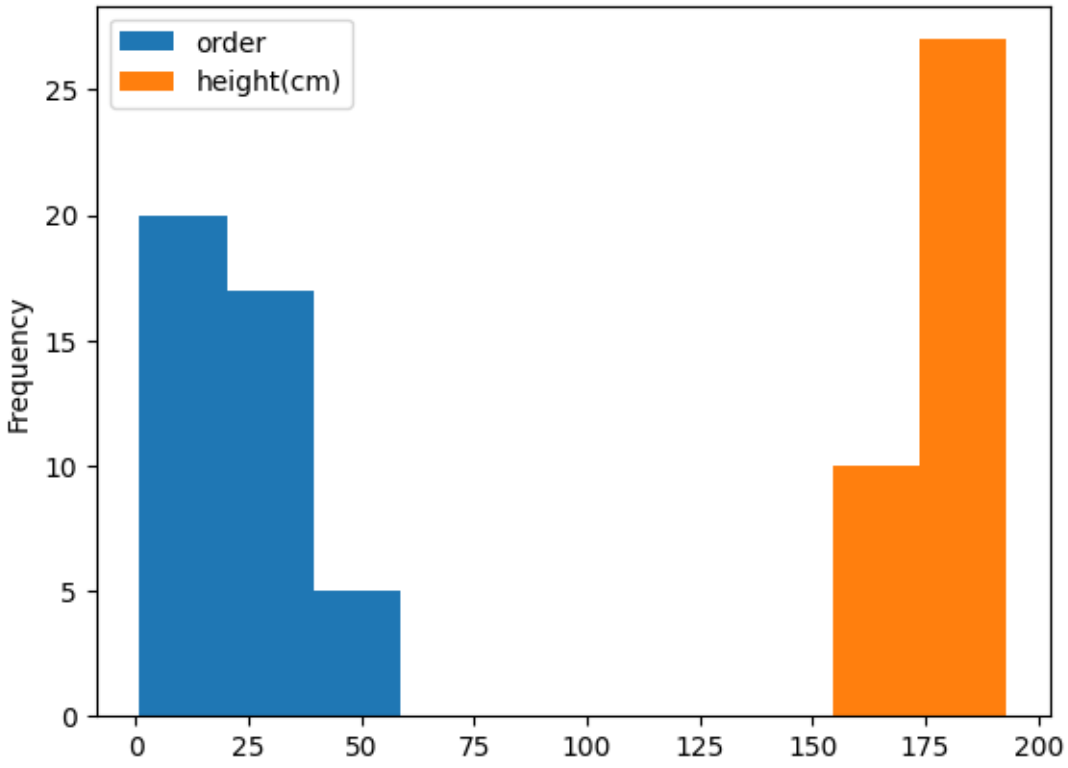
[26]:
```
df['height(cm)'].plot()
```

[26]:
```
<Axes: >
```



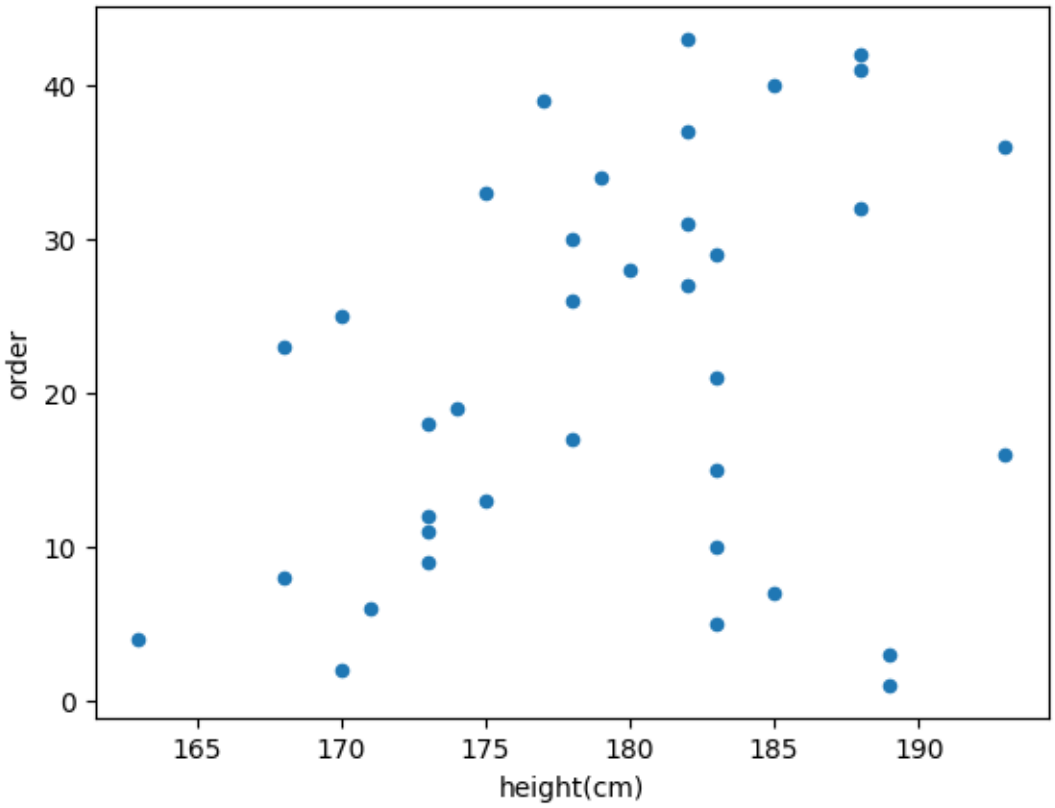
[27]:
```
df.plot.hist()
```

[27]:
```
<Axes: ylabel='Frequency'>
```



[28]:
```
df.plot.scatter(x='height(cm)',y='order')
```

[28]:
```
<Axes: xlabel='height(cm)', ylabel='order'>
```



[29]:
```
print(df.ffill())
```

	order	name	height(cm)
0	16	Abraham Lincoln	193.0
1	7	Andrew Jackson	185.0
2	17	Andrew Johnson	178.0
3	44	Barack Obama	178.0
4	23	Benjamin Harrison	168.0
5	42	Bill Clinton	188.0
6	30	Calvin Coolidge	178.0
7	21	Chester A. Arthur	183.0
8	34	Dwight D. Eisenhower	179.0
9	32	Franklin D. Roosevelt	188.0
10	14	Franklin Pierce	188.0
11	41	George H. W. Bush	188.0
12	43	George W. Bush	182.0
13	1	George Washington	189.0
14	38	Gerald Ford	189.0
15	33	Harry S. Truman	175.0
16	31	Herbert Hoover	182.0
17	20	James A. Garfield	182.0

```
18     15          James Buchanan       183.0
19     11           James K. Polk       173.0
20      4           James Madison       163.0
21      5            James Monroe       183.0
22     39           Jimmy Carter       177.0
23      2             John Adams       170.0
24     35         John F. Kennedy       170.0
25      6       John Quincy Adams       171.0
26     10             John Tyler       183.0
27     36       Lyndon B. Johnson       193.0
28      8        Martin Van Buren       168.0
29     13        Millard Fillmore       175.0
30     37           Richard Nixon       182.0
31     40           Ronald Reagan       185.0
32     19     Rutherford B. Hayes       174.0
33     26      Theodore Roosevelt       178.0
34      3        Thomas Jefferson       189.0
35     18        Ulysses S. Grant       173.0
36     29       Warren G. Harding       183.0
37      9  William Henry Harrison       173.0
38     27     William Howard Taft       182.0
39     25        William McKinley       170.0
40     28          Woodrow Wilson       180.0
41     12          Zachary Taylor       173.0
```

[30]:
```
print(df.bfill())
```

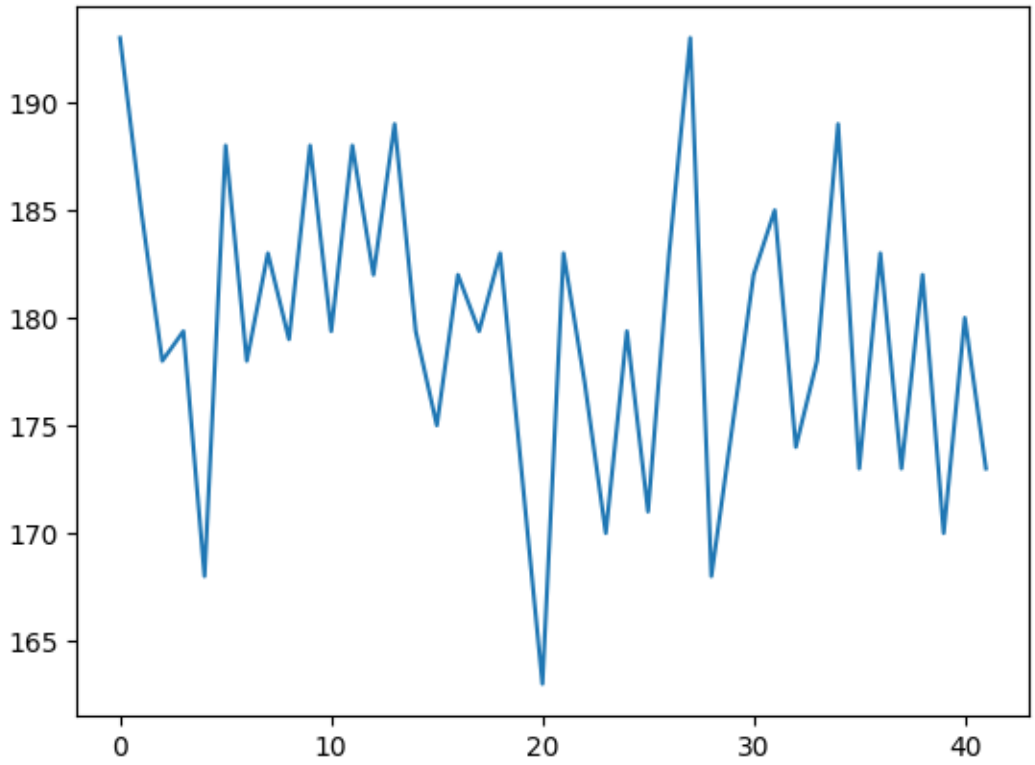
```
    order                    name  height(cm)
0      16         Abraham Lincoln       193.0
1       7          Andrew Jackson       185.0
2      17          Andrew Johnson       178.0
3      44            Barack Obama       168.0
4      23       Benjamin Harrison       168.0
5      42            Bill Clinton       188.0
6      30         Calvin Coolidge       178.0
7      21       Chester A. Arthur       183.0
8      34    Dwight D. Eisenhower       179.0
9      32   Franklin D. Roosevelt       188.0
10     14         Franklin Pierce       188.0
11     41       George H. W. Bush       188.0
12     43          George W. Bush       182.0
13      1       George Washington       189.0
14     38             Gerald Ford       175.0
15     33         Harry S. Truman       175.0
16     31          Herbert Hoover       182.0
17     20       James A. Garfield       183.0
18     15          James Buchanan       183.0
19     11           James K. Polk       173.0
```

20	4	James Madison	163.0
21	5	James Monroe	183.0
22	39	Jimmy Carter	177.0
23	2	John Adams	170.0
24	35	John F. Kennedy	171.0
25	6	John Quincy Adams	171.0
26	10	John Tyler	183.0
27	36	Lyndon B. Johnson	193.0
28	8	Martin Van Buren	168.0
29	13	Millard Fillmore	175.0
30	37	Richard Nixon	182.0
31	40	Ronald Reagan	185.0
32	19	Rutherford B. Hayes	174.0
33	26	Theodore Roosevelt	178.0
34	3	Thomas Jefferson	189.0
35	18	Ulysses S. Grant	173.0
36	29	Warren G. Harding	183.0
37	9	William Henry Harrison	173.0
38	27	William Howard Taft	182.0
39	25	William McKinley	170.0
40	28	Woodrow Wilson	180.0
41	12	Zachary Taylor	173.0

[31]:
```
b1=df["height(cm)"].mean()
df=df.fillna(b1)
```

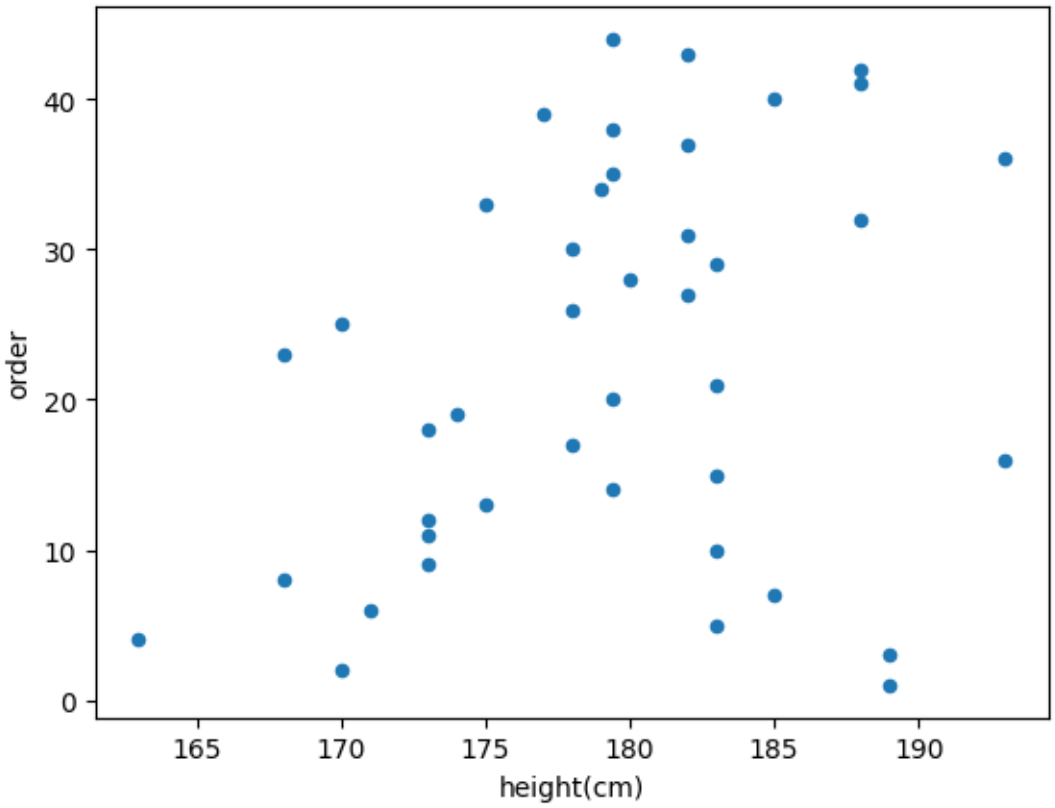
[32]:
```
df['height(cm)'].plot()
```

[32]: <Axes: >



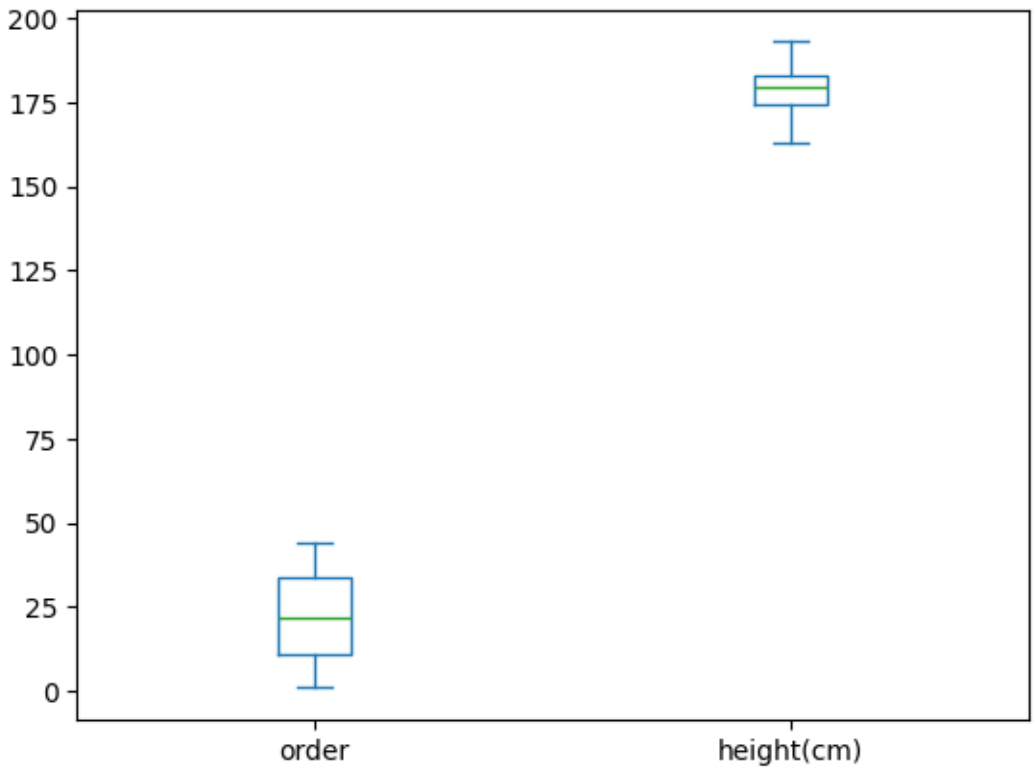
[33]:
```
df.plot.scatter(x='height(cm)',y='order')
```

[33]:
```
<Axes: xlabel='height(cm)', ylabel='order'>
```



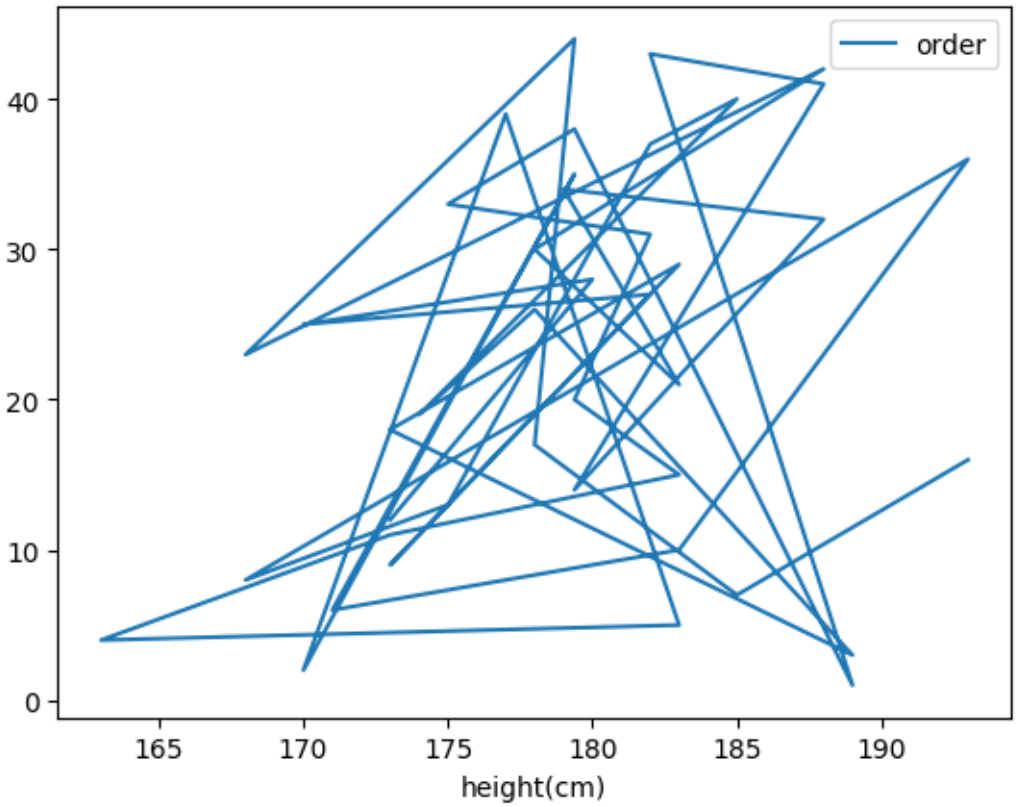
[34]:
```
df.plot.box()
```

[34]: <Axes: >



[35]:
```
df.plot.line(x='height(cm)')
```

[35]:
```
<Axes: xlabel='height(cm)'>
```


order
40
30
20
10
0
165
170
175
180
185
190
height(cm)